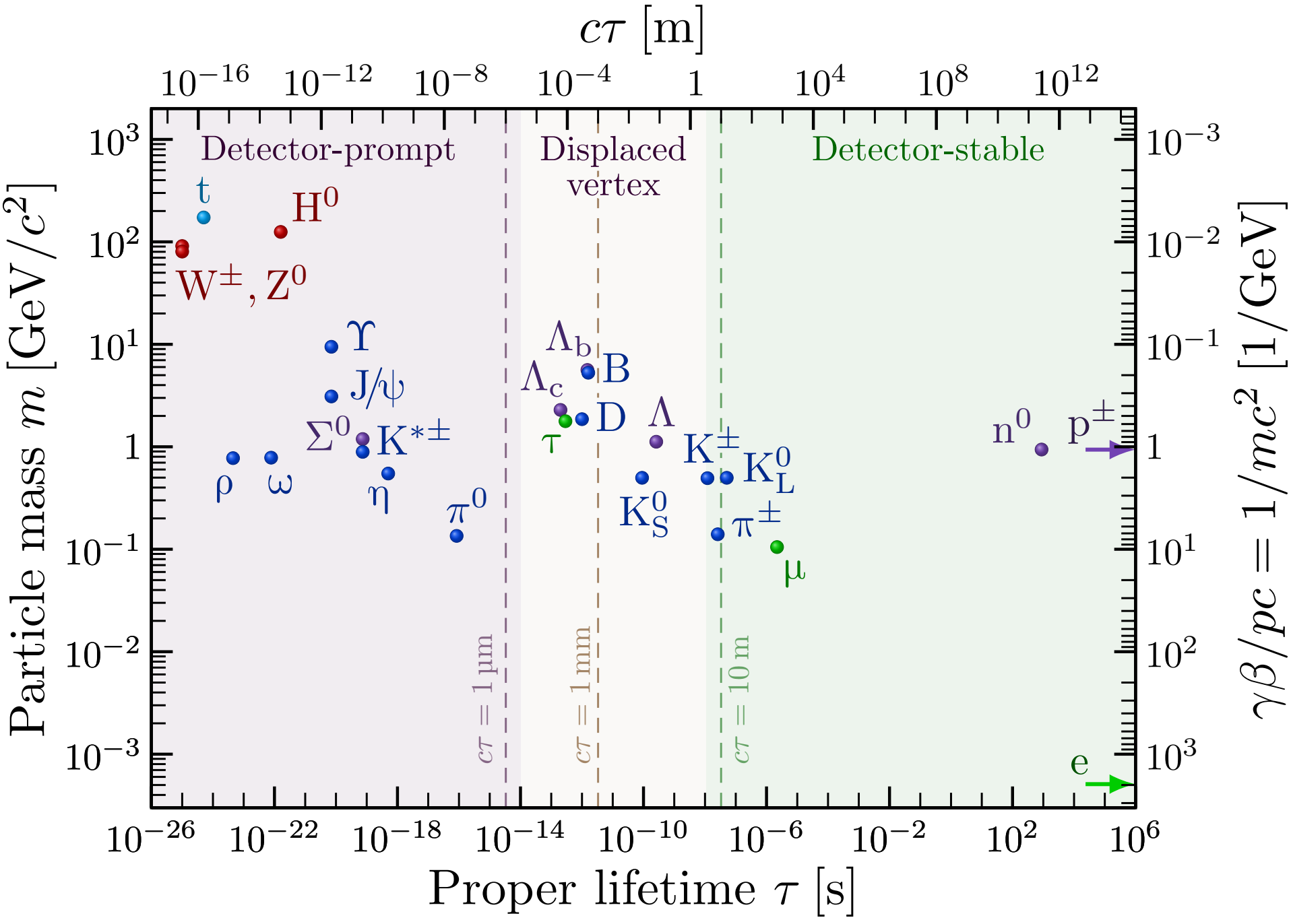
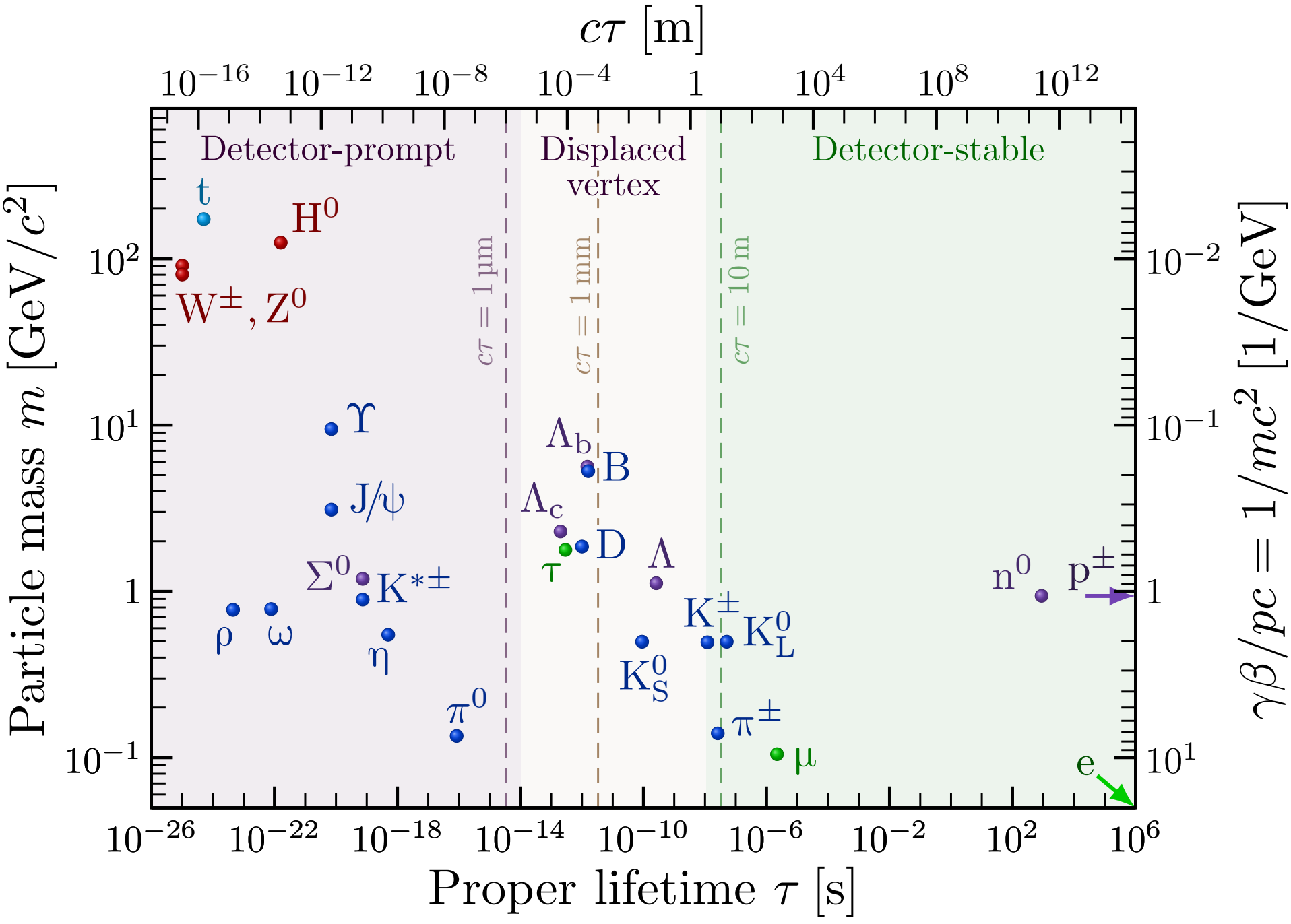
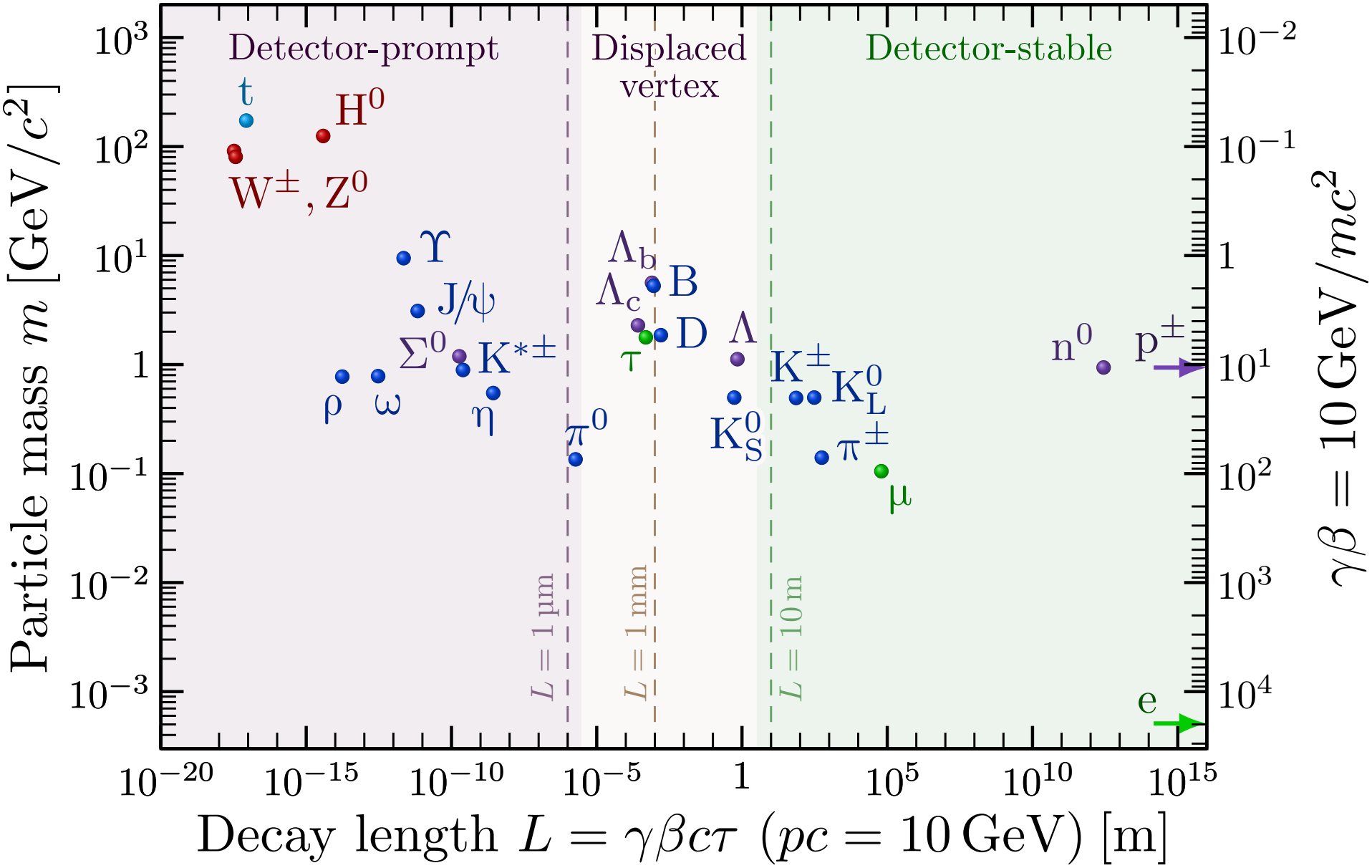
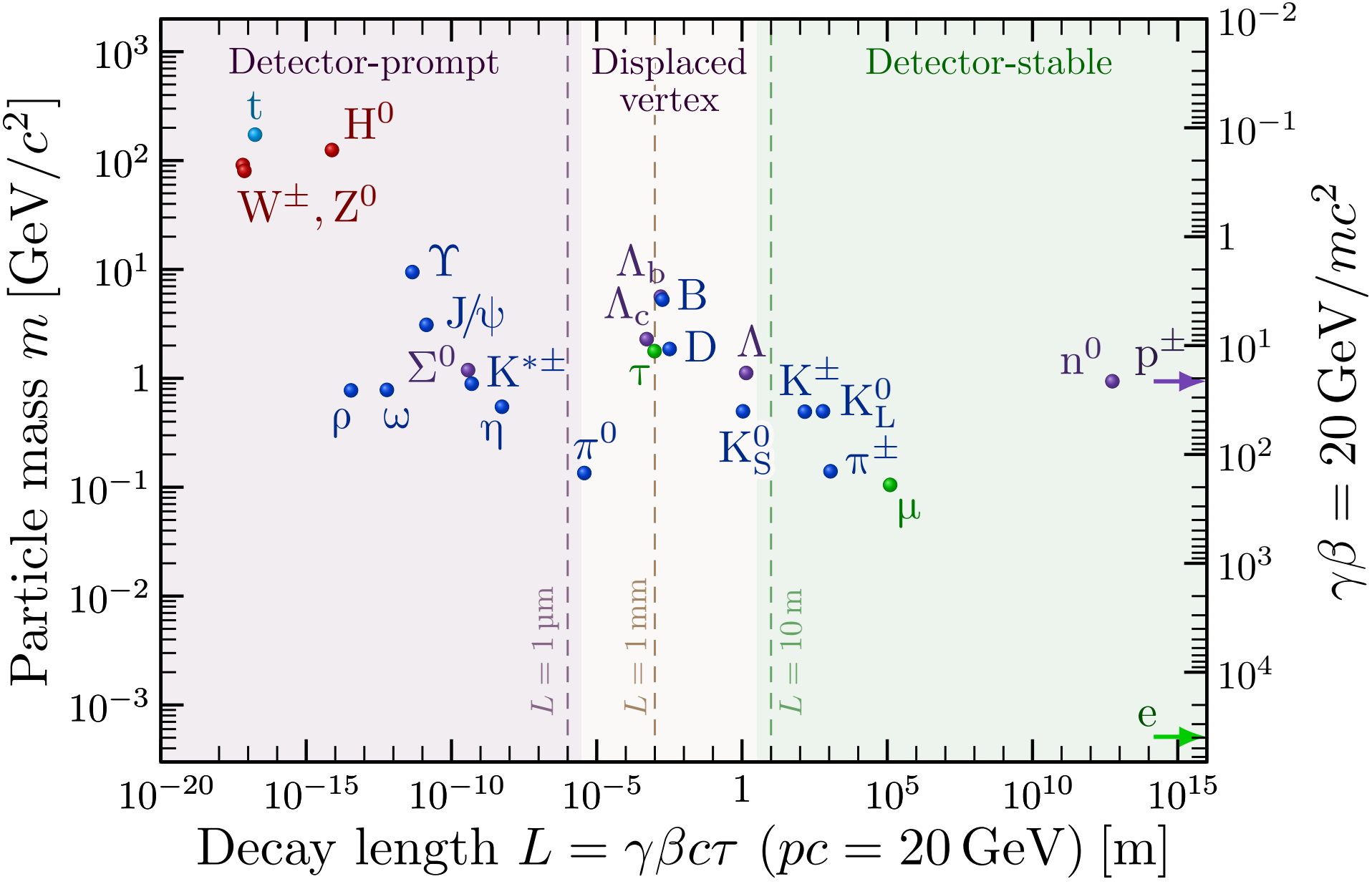
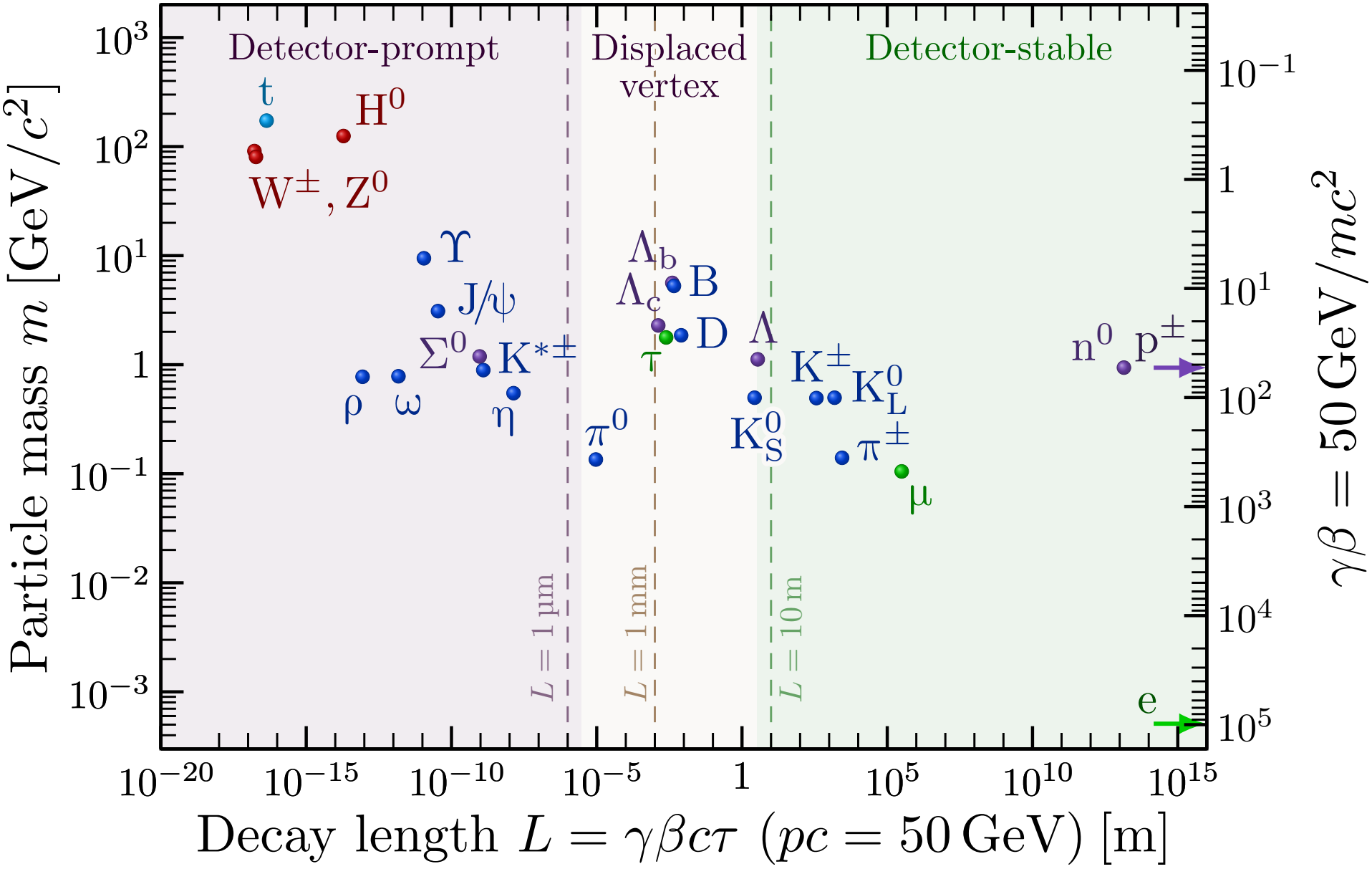
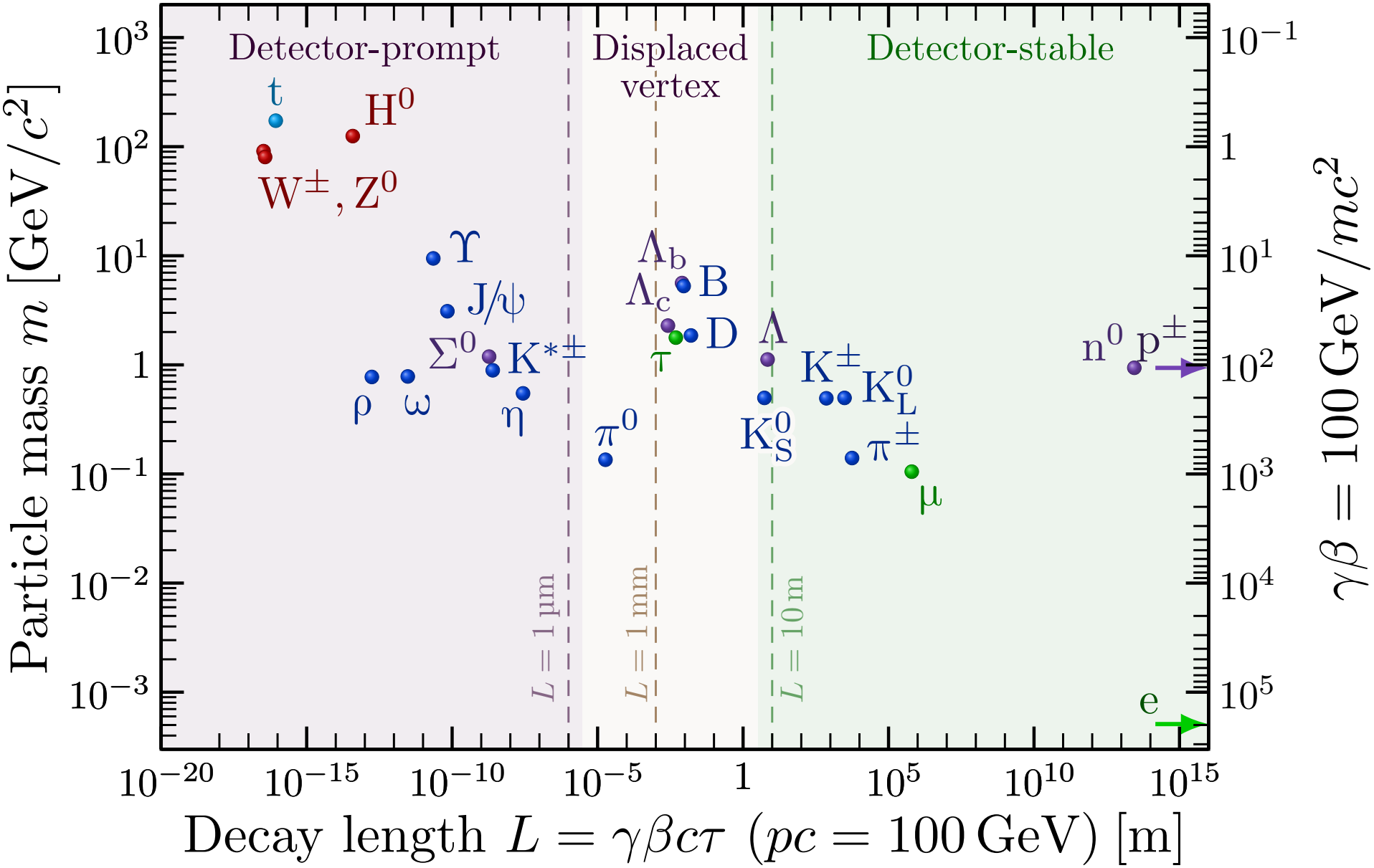

$c\tau$ [m]
10^{-16} 10^{-12} 10^{-8} 10^{-4} 1 10^{4} 10^{8} 10^{12}
Detector-prompt
Displaced vertex
Detector-stable
t
H^0
$W^{\pm}, Z^0$
Υ
J/ψ
Σ^0
$K^{*\pm}$
ρ
ω
η
π^0
Λ_b
B
Λ_c
D
τ
Λ
$K^{\pm}$
K^0_L
K^0_S
$\pi^{\pm}$
μ
n^0
$p^{\pm}$
e
$c\tau = 1\,\mu$m
$c\tau = 1\,$mm
$c\tau = 10\,$m
Particle mass m [GeV/c^2]
10^{3} 10^{2} 10^{1} 1 10^{-1} 10^{-2} 10^{-3}
$\gamma\beta/pc = 1/mc^2$ [1/GeV]
10^{-3} 10^{-2} 10^{-1} 1 10^{1} 10^{2} 10^{3}
10^{-26} 10^{-22} 10^{-18} 10^{-14} 10^{-10} 10^{-6} 10^{-2} 10^{2} 10^{6}
Proper lifetime τ [s]

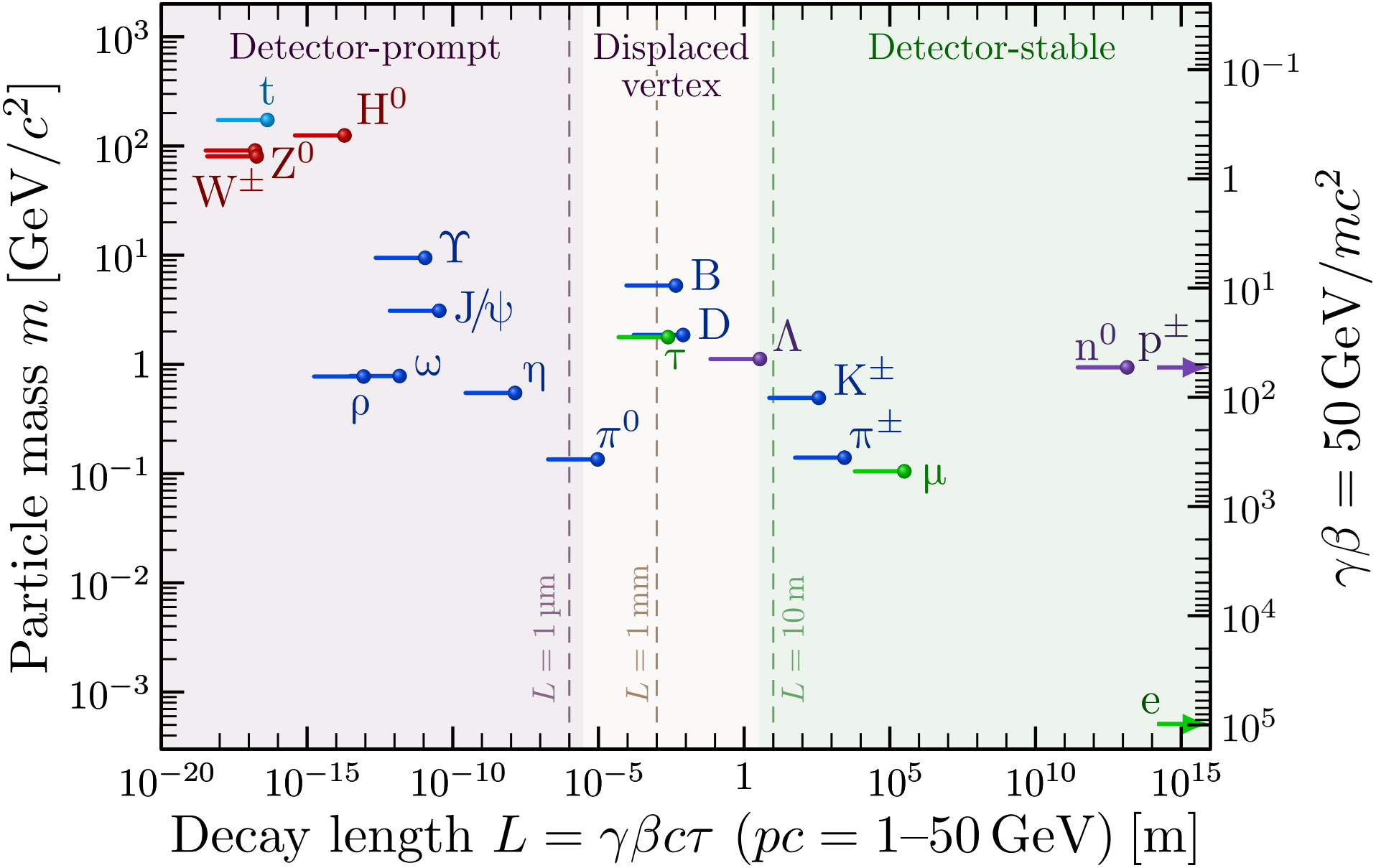

$c\tau$ [m]
10^{-16} 10^{-12} 10^{-8} 10^{-4} 1 10^{4} 10^{8} 10^{12}
Detector-prompt
Displaced vertex
Detector-stable
$c\tau = 1\,\mu\text{m}$
$c\tau = 1\,\text{mm}$
$c\tau = 10\,\text{m}$
t
H^0
$W^\pm, Z^0$
Υ
J/ψ
Σ^0
$K^{*\pm}$
ρ
ω
η
π^0
Λ_b
B
Λ_c
D
τ
Λ
K^0_S
$K^\pm$
K^0_L
$\pi^\pm$
μ
n^0
$p^\pm$
e
Particle mass m [GeV/c^2]
10^2 10^1 1 10^{-1}
$\gamma\beta/pc = 1/mc^2$ [1/GeV]
10^{-2} 10^{-1} 1 10^1
10^{-26} 10^{-22} 10^{-18} 10^{-14} 10^{-10} 10^{-6} 10^{-2} 10^{2} 10^{6}
Proper lifetime τ [s]


Detector-prompt
Displaced vertex
Detector-stable
t
H^0
W$^\pm$, Z^0
Υ
J/ψ
Σ^0
K$^{*\pm}$
ρ
ω
η
Λ_b
Λ_c
B
D
τ
Λ
π^0
K^0_S
K$^\pm$
K^0_L
$\pi^\pm$
μ
n^0
p$^\pm$
e
$L = 1\,\mu$m
$L = 1\,$mm
$L = 10\,$m
Particle mass m [GeV/c^2]
$\gamma\beta = 10\,\mathrm{GeV}/mc^2$
Decay length $L = \gamma\beta c\tau$ ($pc = 10\,$GeV) [m]


Detector-prompt
Displaced vertex
Detector-stable
t
H^0
W$^\pm$, Z^0
Υ
J/ψ
Σ^0
K$^{*\pm}$
ρ
ω
η
Λ_b
Λ_c
B
D
τ
π^0
Λ
K$^\pm$
K^0_L
K^0_S
$\pi^\pm$
μ
n^0
p$^\pm$
e
$L = 1\,\mu$m
$L = 1\,$mm
$L = 10\,$m
Particle mass m [GeV/c^2]
$\gamma\beta = 20\,\mathrm{GeV}/mc^2$
Decay length $L = \gamma\beta c\tau$ ($pc = 20\,$GeV) [m]
10^{-3}
10^{-2}
10^{-1}
1
10^1
10^2
10^3
10^4
10^{-2}
10^{-1}
1
10^1
10^2
10^3
10^4
10^{-20}
10^{-15}
10^{-10}
10^{-5}
1
10^5
10^{10}
10^{15}


Detector-prompt
Displaced vertex
Detector-stable
t
H^0
W$^\pm$, Z^0
Υ
J/ψ
Σ^0
K$^{*\pm}$
ρ
ω
η
Λ_b
Λ_c
B
D
τ
π^0
Λ
K^0_S
K$^\pm$
K^0_L
$\pi^\pm$
μ
n^0
p$^\pm$
e
$L = 1\,\mu$m
$L = 1\,$mm
$L = 10\,$m
Particle mass m [GeV/c^2]
$\gamma\beta = 50\,\mathrm{GeV}/mc^2$
Decay length $L = \gamma\beta c\tau$ ($pc = 50\,$GeV) [m]


Detector-prompt
Displaced vertex
Detector-stable
t
H^0
W$^\pm$, Z^0
Υ
J/ψ
Σ^0
K$^{*\pm}$
ρ
ω
η
Λ_b
Λ_c
B
D
τ
π^0
Λ
K^0_S
K$^\pm$
K^0_L
$\pi^\pm$
μ
n^0
p$^\pm$
e
$L = 1\,\mu$m
$L = 1\,$mm
$L = 10\,$m
Particle mass m [GeV/c^2]
$\gamma\beta = 100\,\mathrm{GeV}/mc^2$
Decay length $L = \gamma\beta c\tau$ ($pc = 100\,$GeV) [m]


Detector-prompt
Displaced vertex
Detector-stable
t
H^0
Z^0
W$^\pm$
Υ
J/ψ
ω
ρ
η
B
D
τ
Λ
π^0
K$^\pm$
$\pi^\pm$
μ
n^0
p$^\pm$
e
$L = 1\,\mu$m
$L = 1\,$mm
$L = 10\,$m
Particle mass m [GeV/c^2]
$\gamma\beta = 50\,\mathrm{GeV}/mc^2$
Decay length $L = \gamma\beta c\tau$ ($pc = 1$–$50\,$GeV) [m]